Done

Puzzle: Break a Hashing Function

In this assignment you will write a simple hashing function that uses addition and multiplication and then find a pair of strings that will return the same hash value for different strings (i.e you will cause a Hash Collision (https://en.wikipedia.org/wiki/Collision_(computer_science))).

The algorithm uses multiplication based on the position of a letter in the hash to avoid a hash collision when two letters are transposed like in 'ABCDE' and 'ABDCE'. Your strings need to be at least three characters long and no more than 10 characters long.

Enter your two strings below and check if they cause a collision:

String 1... String 2...

Check for Collision

Here is the code that computes your hash:

```
while True:
    txt = input("Enter a string: ")
    if len(txt) < 1 : break

    hv = 0
    pos = 0
    for let in txt:
        pos = ( pos % 3 ) + 1
        hv = (hv + (pos * ord(let))) % 1000000
        print(let, pos, ord(let), hv)

    print(hv, txt)
```

For simplicity we will only use upper and lower case ASCII letters in our text strings.

Done

```
Enter a string: ABCDE
A 1 65 65
B 2 66 197
C 3 67 398
D 1 68 466
E 2 69 604
604 ABCDE

Enter a string: BACDE
B 1 66 66
A 2 65 196
C 3 67 397
D 1 68 465
E 2 69 603
603 BACDE
```

Note: This is a puzzle assignment. The answer is ultimately very simple. All you need to do is **completely** understand two lines of Python code. It is easy to get the solution from another student or perhaps ask StackOverflow. But if you do that you deprive yourself of the satisfaction of solving the puzzle and the learning moment that happens when you look at something, break it down, figure it out, and solve it on your own. If you feel stuck - take a break then come back.

Here is a video to watch so you can take a break from working on this. Bruce Schneier: The Security Mindset (https://www.youtube.com/watch?v=eZNzMKS7zjo).